

Answer ALL questions.

Write your answers in the spaces provided.

- 1** Animal and plant cells can be observed using a light microscope.

The structure of human cheek cells was investigated.

- (a) Describe a safe method to prepare and examine the structure of human cheek cells.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



(b) The photograph shows three human cheek cells, seen using a light microscope.



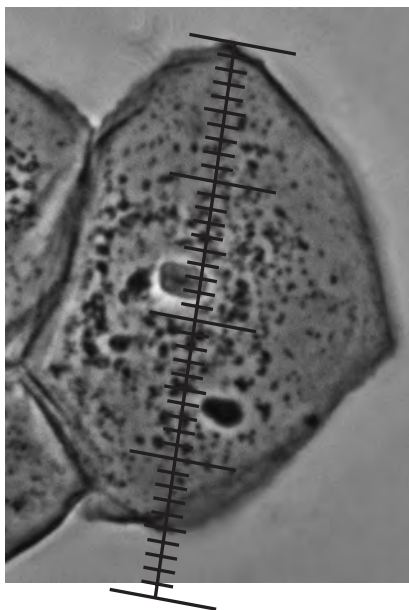
(Source: © BIOPHOTO ASSOCIATES/SCIENCE PHOTO LIBRARY)

- (i) Draw the cell labeled W in the photograph. Label two structures on your drawing. (2)



P 6 6 1 5 5 A 0 3 2 0

- (ii) The photograph shows an eyepiece graticule over the widest part of this human cheek cell.



(Source: © BIOPHOTO ASSOCIATES/SCIENCE PHOTO LIBRARY)

Each of the smallest divisions on the graticule is 3×10^{-6} m.

Determine the largest width of the cell, using the graticule.

Give your answer in micrometres (μm).

(2)

Answer μm



- (c) Human red blood cells contain a large number of haemoglobin molecules.

Each haemoglobin molecule is a sphere with a diameter of 5 nm.

- (i) Calculate the volume of a haemoglobin molecule.

The equation for the volume of a sphere (V) is

$$V = \frac{4}{3}\pi r^3$$

(1)

Answer nm³

- (ii) A human red blood cell has a volume of 80 μm³.

Calculate how many times larger the volume of a red blood cell is than the volume of a haemoglobin molecule.

Give your answer in standard form.

(2)

Answer



(iii) State **two** differences you would observe between a plant cell and a human red blood cell when using a light microscope.

(2)

1

.....

.....

2

.....

.....

(Total for Question 1 = 13 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

